Proposed-Project Report: DarkWatch XAI

An Explainable, Multi-Modal Cyber Threat Intelligence System for Underground & Anonymous Network Environments

1. Problem Statement

The rapid evolution of cybercrime ecosystems on dark web hidden services and underground forums poses a critical challenge for modern Security Operations Centers (SOCs) 1-3. Cyber threat intelligence (CTI) gathered from these platforms provides crucial early-warning signals regarding emerging zero-day vulnerabilities, stolen credential dumps, ransomware leak claims, and threat actor tradecraft 3-5. However, operationalizing underground threat intelligence faces four severe bottlenecks:

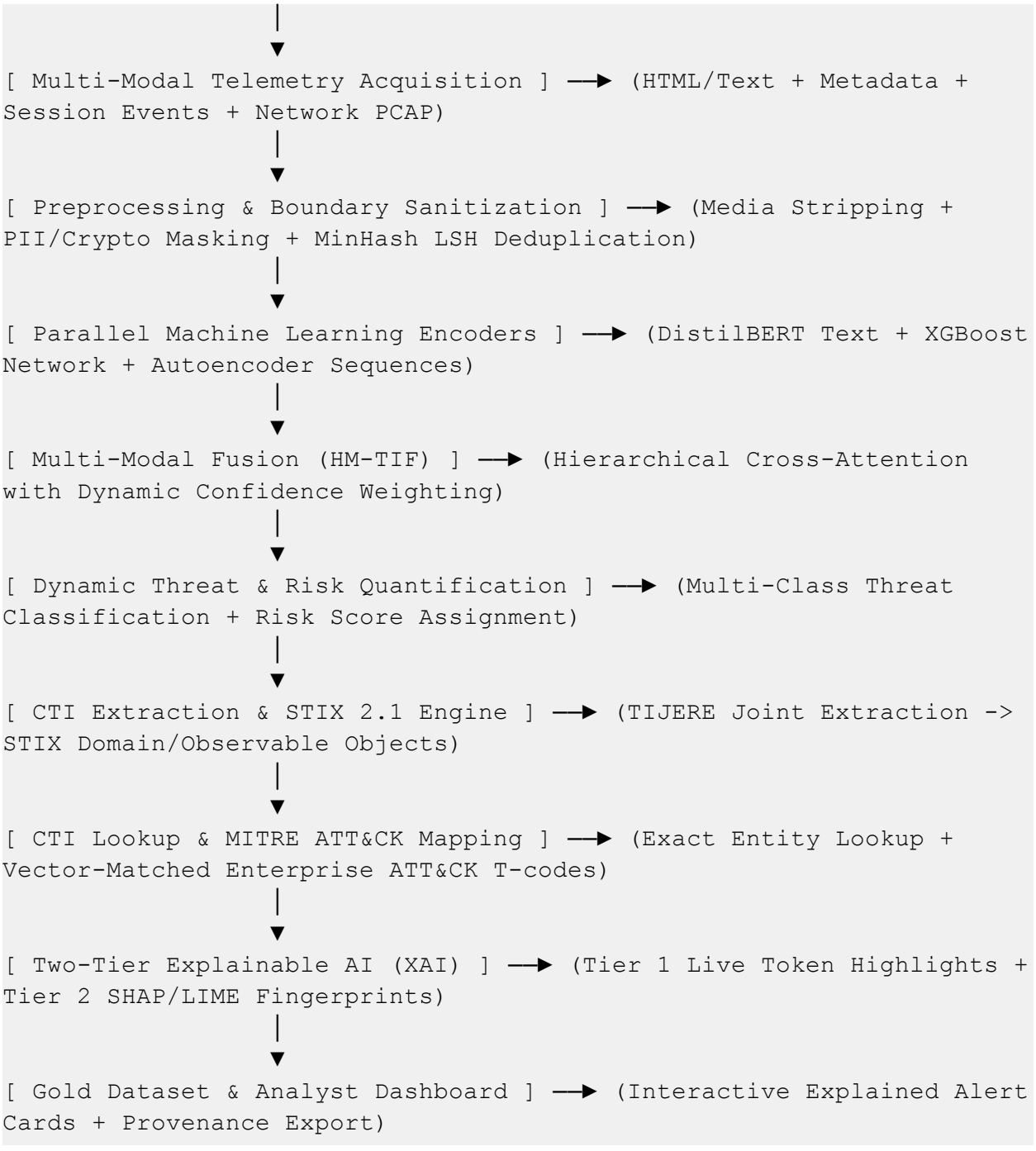
1. **High Environmental Volatility & Redundancy**: Dark web hidden services exhibit extreme availability decay—up to 50% of onion services deactivate within 24 hours of discovery, and only 2.8% to 30% remain reachable over extended periods 6, 7. Furthermore, pervasive content duplication, phishing clones, and mirror sites account for 20% to over 90% of observed dark web pages 8-10.
2. **Unstructured & Heterogeneous Data Sources**: Underground CTI is scattered across unstructured forum posts, paste repositories, encrypted chats, and leak sites, characterized by informal jargon, multilingual posts, and obfuscated handles 11-14.
3. **Analyst Alert Fatigue & Unscalable Manual Review**: Security teams are overwhelmed by thousands of daily threat notifications 15. Manual triage, entity extraction, and cross-referencing against frameworks like MITRE ATT&CK are labor-intensive, error-prone, and introduce significant response delays 12, 14, 16.
4. **Limitations of Existing Datasets & Single-Modality Detection**: Benchmark datasets (such as CoDA 17, DUTA-10K 18, CrimeBB 19, or CICDarknet2020 20) focus either on static text in isolation or on network packet captures in isolation 21, 22. Existing architectures assume naturally aligned, complete multi-modal data streams—a condition that rarely exists in real-world SOC operations where access barriers (CAPTCHAs, 503 DDoS protection walls, dead onion circuits) frequently cause individual data channels to drop out 23, 24.

To address these challenges, there is an urgent scientific need for an **automated, multi-modal, and explainable CTI system** capable of correlating non-aligned threat signals, extracting structured knowledge, mapping adversarial behaviors, and providing human-understandable risk explanations 21, 23, 25.

2. Proposed Approach & End-to-End Pipeline

DarkWatch XAI addresses these limitations through an integrated, multi-stage processing pipeline that spans from passive observation to explainable alert delivery:

```
[ Authorized Service Observation ] ⟶ (Whonix Dual-VM Isolation / Sandboxed Service Testbed)
```



3. Observation & Experimental Environment

To evaluate system performance safely and reproducibly without legal or ethical exposure, DarkWatch employs a **dual-layer collection and experimental architecture**:

- **Whonix Isolation Boundary**: Real-world dark web crawls execute inside **Whonix-Workstation**, leveraging a dual-VM topology where all network traffic is forcibly routed through the **Whonix-Gateway** proxy (socks5h://10.152.152.10:9050) 26, 27. This enforces a fail-closed boundary, preventing accidental direct IP or DNS leaks 26, 28.

Non-text media (images, executables, scripts) are stripped at download, and transient identifiers ([ID_CRYPTO], [ID_ONION_URL], [ID_IP_ADDRESS]) are masked prior to storage 29.

- **Containerized Emulated Service Testbed**: To simulate underground service archetypes under controlled conditions, DarkWatch deploys a suite of lightweight Docker services:
- *Threaded Discussion Forum*: Emulates communities like BreachForums 30 or CrimeBB 19 (generating technical text, user metadata, and reply structures).
- *Marketplace / Listing Portal*: Emulates darknet markets (generating structured product text, price quotes, and wallet references) 11, 31.
- *Pastebin / Code Sharing*: Emulates paste sites (generating raw exploit code snippets, configuration files, and CVE mentions) 3.
- *Ransomware Leak Portal*: Emulates extortion sites (generating victim notices and file hash lists without hosting malicious binaries) 32.
- *Access Barrier Interstitial*: Emulates Cloudflare or DDoS-Guard challenge walls (returning HTTP 403/503 status codes and low-density challenge HTML) 23.

Experimental Variable Injection

Rather than static features, transport and access controls are treated as independent variables:

- **Transport Modes**: Tor multi-hop routing (socks5h) vs. VPN vs. Direct/clearnet (generating distinct network flow features as in CICDarknet2020) 20, 33.
- **Access Control Barriers**: Direct access vs. 2-step CAPTCHA challenge vs. 2FA/session gated (simulating AUTOMATED_ACCESS_BLOCKED states to benchmark missing-modality AI resilience) 23, 34.
- **Content Dynamics**: Static baseline vs. rapidly changing threads vs. near-duplicate mirror pages (evaluated via MinHash LSH signatures) 8, 9.

4. Dataset Structure & Session Linkage

The DarkWatch Gold Dataset stores correlated **evidence records** \\(\mathcal{E} = (T_{\text{text}}, N_{\text{network}}, S_{\text{temporal}})\\) with end-to-end provenance:

- **HTML & Text Layer (\\(T\\))**: Raw HTML, extracted clean body text (via Trafilatura), DOM structural metadata, form types, and masked tokens.
- **Network Telemetry Layer (\\(N\\))**: 80 flow-level features extracted via CICFlowMeter (flow duration, packet size variance, inter-arrival times, TCP flags) 20.
- **Session & Behavioral Layer (\\(S\\))**: Navigation event sequences, session cookie timestamps, request frequencies, and click-stream interaction paths.
- **Temporal Dynamics**: Versioned text diffs and MinHash Jaccard similarity scores tracking content changes over time 8.
- **CTI & MITRE Annotations**: Extracted STIX 2.1 entities (CVEs, file hashes, malware families, tools) and mapped Enterprise ATT&CK T-codes 35-37.
- **Provenance & Governance**: schema_version, collector_engine, sanitizer_engine, raw_html_sha256, clean_text_sha256, timestamps, and multi-tier annotation history (AUTOMATED, WEAK_SUPERVISION, HUMAN_VERIFIED).

Session & Event Linkage

Multiple observations within the same browsing session or attack campaign are linked as a **Directed Heterogeneous Information Graph**:\\\text{Session State} \longrightarrow \text{Verification/CAPTCHA} \longrightarrow \text{Login Event} \longrightarrow \text{Board Navigation} \longrightarrow \text{Thread View} \longrightarrow \text{Post/Reply} \longrightarrow \text{Content Delta}\\This graph structure mirrors the STIX 2.1 relationship schema (observed-data \\(\rightarrow\\) indicator \\(\rightarrow\\) attack-pattern) 36, 37 and user representation models like **URM4DMU** 38.

5. AI-Trained vs. Deterministic CTI Processing

To ensure scientific clarity, DarkWatch strictly delineates **machine learning-trained components** from **deterministic extraction, knowledge representation, and lookup engines**:

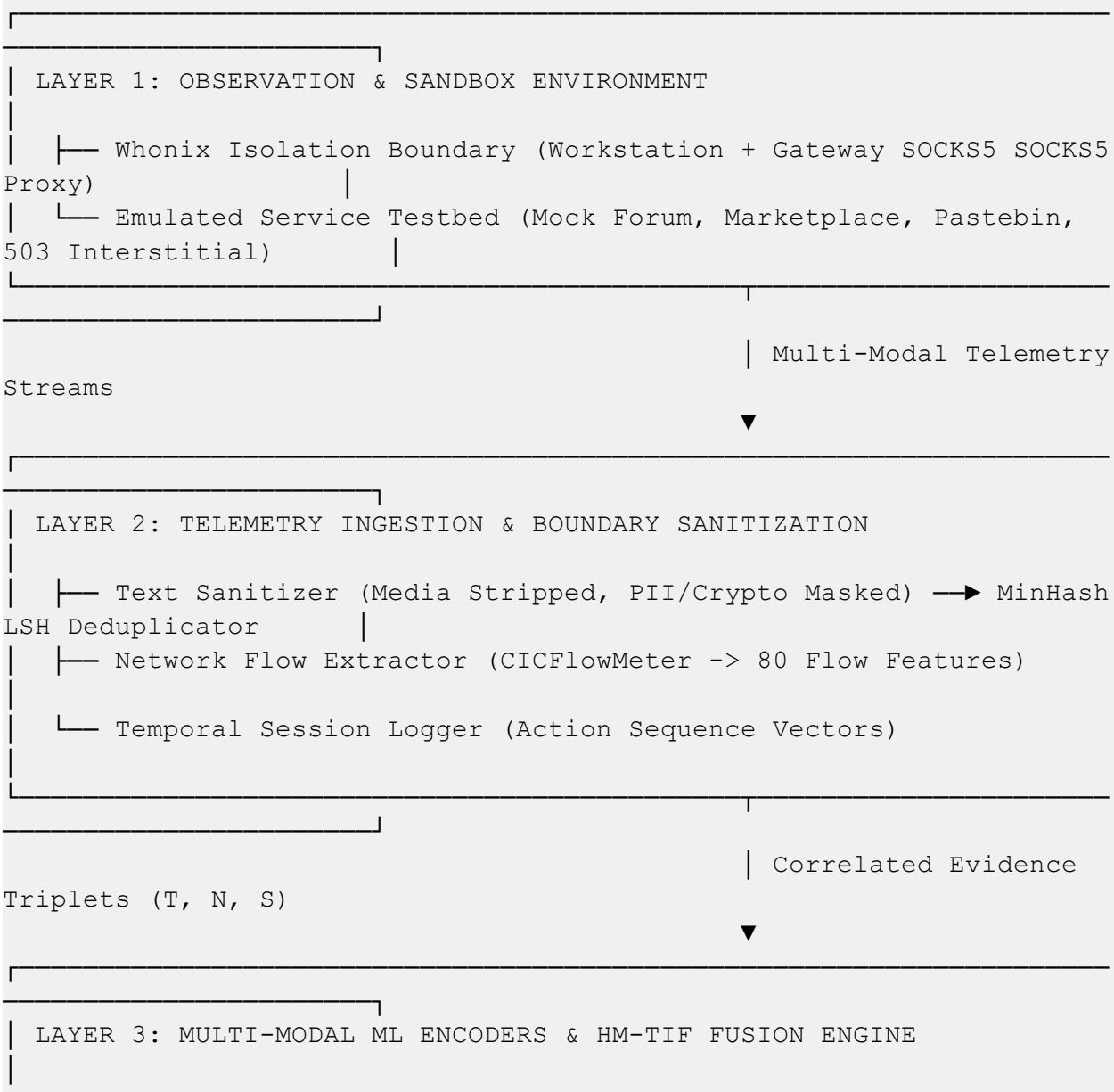
Component,Architecture / Mechanism,Operation Type,Role in DarkWatch
Text Classifier,"Fine-Tuned distilbert-base-uncased / SecureBERT 14, 39",ML-Trained,"Multi-label classification of activity categories (Hacking, Malware, Vulnerability) and threat intent."
Network Classifier,"Multiclass XGBoost / GNN on 80 flow features 20, 21, 33",ML-Trained,"Classifies encrypted traffic categories (browsing, C2, exfiltration) and network anomalies."
Temporal Model,"BiLSTM / Autoencoder sequence models 21, 40",ML-Trained,Models behavioral action frequencies and detects temporal sequence anomalies.
Multi-Modal Fusion,"Hierarchical Multi-Modal Fusion (HM-TIF) with Cross-Attention 21, 23, 41",ML-Trained,Dynamically weights modality contributions based on correlation confidence and data availability.
CTI Entity Extraction,"TIJERE Joint Entity-Relation Model / Regex rules 3, 42",Hybrid (ML + Deterministic),"Extracts technical IoCs (CVEs, IPs, hashes, malware names, tools) from unstructured text."
STIX 2.1 Engine,"Python stix2 library & JSON schema builders 36, 37, 43",Deterministic Knowledge Engine,"Generates standardized STIX Domain Objects (SDOs), Cyber-Observables (SCOs), and Relationship Objects (SROs)."
CTI Correlation,"SQLite / Local CTI Repository Exact Lookups 8, 43",Deterministic Lookup,Queries local historical records to link extracted indicators to past threat actor campaigns.
MITRE ATT&CK Mapper,"Offline STIX KB Lookup + VTT-LLM Semantic Matcher 36, 44, 45",Hybrid (Lookup + Vector Match),"Resolves extracted attack patterns to official Enterprise ATT&CK T-codes (T1190, T1486) and mitigations."
Explainable AI (XAI),"Post-hoc SHAP, LIME, and Cross-Attention Weights 25, 46-48",Post-Hoc Attribution Engine,Renders Tier 1 attention highlights and Tier 2 feature attribution bar charts on the dashboard.

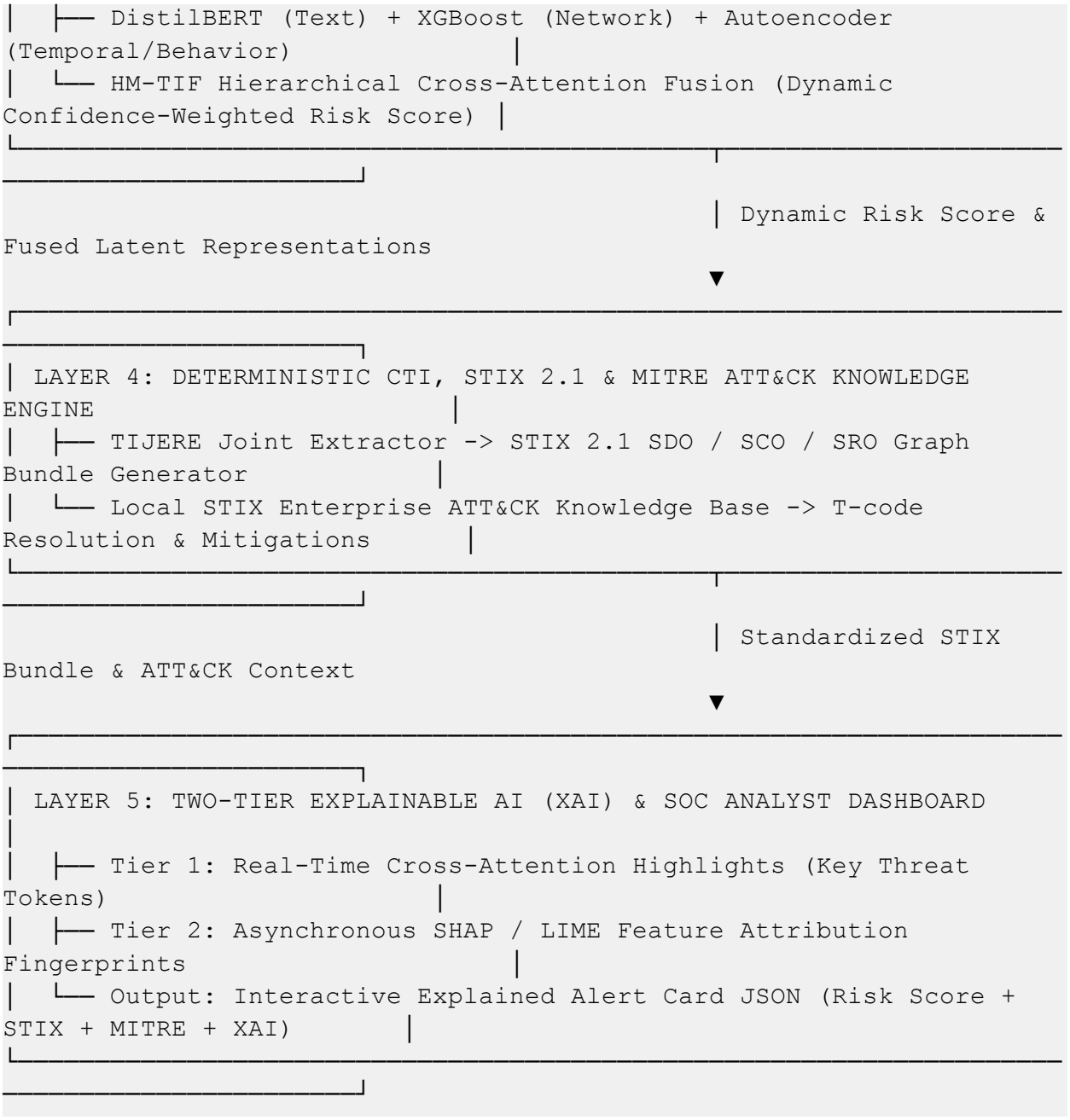
6. Annotation and Training Strategy

DarkWatch implements a **dual-tier annotation governance framework**:

- **Tier 1: WEAK_SUPERVISION (Candidate Generation)**:
- WeakSupervisionEngine applies keyword anchors, regex rules, and CTI entity density to generate candidate labels for Activity, Intent, and Severity.
- Weak labels are flagged as candidate hypotheses, **never as ground truth**.
- **Tier 2: HUMAN_VERIFIED (Adjudicated Gold Benchmark)**:
- Analysts review: (a) disputed candidate labels, (b) high-severity CVE/ransomware records, and (c) a stratified random sample reserved for testing 35, 49.
- Adjudication calculates inter-annotator agreement using **Cohen's Kappa (\\(\kappa_w\\))** (targeting \\(\kappa_w \ge 0.65\\)) 35, 49.
- Approved records are promoted to HUMAN_VERIFIED status in annotation_store.py.

7. Complete System Architecture





8. Research Questions and Objectives

Primary Research Questions

- **RQ1 (Access-Barrier Resilience)**: How severely does threat detection performance degrade when access barriers (CAPTCHAs, 503 challenge walls) cause textual CTI channels to drop out, and can multi-modal cross-attention fusion maintain detection accuracy using network and temporal telemetry alone 23, 34?
- **RQ2 (Fusion Superiority)**: Does hierarchical multi-modal cross-attention fusion (HM-TIF) achieve statistically significant improvements in Macro-F1 score and false

positive reduction compared to unimodal baselines and early/late fusion concatenation 21, 23, 50?

- **RQ3 (XAI Explanation Stability)**: How consistent and interpretable are SHAP and LIME feature attributions across changing dark web content dynamics, transport wrappers (Tor vs. VPN), and model architectures 47, 51, 52?
- **RQ4 (Deduplication Bias Mitigation)**: What is the quantitative impact of MinHash LSH deduplication on eliminating training data leakage and reducing classification bias across mirrored dark web pages 8, 9?

Primary Research Objectives

1. Implement a Whonix-isolated, multi-modal observation and data curation pipeline for dark web threat intelligence.
2. Develop a missing-modality resilient fusion model leveraging hierarchical cross-attention (HM-TIF).
3. Build an automated STIX 2.1 and MITRE ATT&CK enrichment engine that links unstructured threat text to standardized adversarial TTPs and mitigations 36, 37, 43.
4. Render transparent, two-tier Explainable AI alert cards that reduce analyst decision latency and mitigate SOC alert fatigue 25, 53.

9. Evaluation Framework

The system will be evaluated across six comprehensive experimental dimensions:

- **Baseline Model Ladder**:
- *Lexical*: TF-IDF + Logistic Regression / Linear SVM 54, 55.
- *Unimodal LMs*: Vanilla distilbert-base-uncased, bert-base-uncased, SecureBERT, and zero-shot DarkBERT 14, 39, 56.
- *Unimodal Network/Behavior*: Standalone XGBoost, Random Forest, and GNN models 20, 21, 55.
- *Concatenation Fusion*: Early fusion (feature concatenation) and Late fusion (decision-level averaging) 21, 57, 58.
- *Proposed Architecture*: **DarkWatch HM-TIF Cross-Attention Fusion**.
- **Ablation & Missing-Modality Studies**:
- Evaluate detection accuracy, Precision, Recall, and Macro-F1 when individual modalities (Text-only, Network-only, Temporal-only) are systematically dropped or delayed 23, 59.
- **Temporal & Source-Disjoint OOD Splits**:
- Partition training and test sets strictly by **crawl timestamp** (Train = earlier batches, Test = recent batches) and seed domain to evaluate prospective generalization under concept drift 60, 61.
- **CTI & MITRE Mapping Metrics**:
- Evaluate STIX entity extraction (TIJERE) and MITRE ATT&CK technique mapping using Precision, Recall, and Micro/Macro F1 scores against human-adjudicated gold references 35, 42.
- **Explainable AI (XAI) Metrics**:

- Assess explanation **Fidelity** (fidelity of local approximations), **Stability** (consistency under minor input perturbations), **Sparsity** (number of features required for an explanation), and computational latency overhead 62-64.
- **Analyst Usability Study**:
- Measure SOC analyst decision latency, trust rating, and false-positive triage speed when viewing Explained Alert Cards versus raw black-box alerts 25, 53, 65.

10. Research Contributions, Limitations, and Final Summary

Genuine Research Contributions

1. **First Access-Barrier Resilient Multi-Modal CTI Framework**: Demonstrates that multi-modal threat intelligence fusion provides operational benefits (over 30% false-positive reduction) even when text streams are blocked or unaligned 23, 34, 66.
2. **Standardized STIX 2.1 & MITRE ATT&CK Mapping Engine**: Integrates joint entity-relation extraction with Enterprise ATT&CK knowledge bases, converting unstructured dark web chatter into actionable STIX 2.1 graph bundles 3, 36, 37, 42.
3. **Traceable Two-Tier XAI for SOC Workflows**: Combines Tier 1 real-time cross-attention token highlights with Tier 2 post-hoc SHAP/LIME feature fingerprints, eliminating alert fatigue and justifying risk scores 25, 46, 48, 51.
4. **Organic Gold Dataset Curation**: Establishes a deduplicated, multi-dimensional dark web CTI dataset with complete provenance and Cohen's Kappa quality verification 8, 35, 49.

Major Limitations and Methodological Risks

- **Synthetic-to-Real Domain Shift**: Models trained exclusively on mock lab web traffic may overfit to synthetic patterns; requires continuous evaluation against real-world dark web crawls 67.
- **XAI Computational Overhead**: Generating SHAP and LIME explanations for long sequences introduces non-trivial computational latency that must be managed via asynchronous background queues 62, 68, 69.
- **Expert Annotation Bottleneck**: Human verification of CTI entity relationships and ATT&CK mappings requires expert domain knowledge and represents an operational bottleneck 14, 35, 70.

Final Description of DarkWatch XAI

DarkWatch XAI is an **Explainable, Multi-Modal Cyber Threat Intelligence (CTI) Framework** designed to passively observe, extract, correlate, and explain threats across anonymous dark web hidden services and underground communities. By integrating **textual NLP models**, **encrypted network flow classifiers**, and **temporal session autoencoders** through a **hierarchical cross-attention fusion engine (HM-TIF)**, DarkWatch maintains robust threat detection and dynamic risk quantification even when individual data channels are blocked or incomplete. Fused threat predictions are automatically converted into **standardized STIX 2.1 graph bundles**, mapped to **MITRE Enterprise ATT&CK T-codes and mitigations**, and

delivered to SOC analysts via **interactive, two-tier Explainable AI (XAI) alert cards**—delivering transparent, actionable, and trustworthy threat intelligence at scale.